

Assignment Lists in Python

1. Write a Python program to find the sum of all elements in the list `[10, 20, 30, 40, 50]`.
2. Write a Python program to display the odd and even elements from the list `[10, 23, 11, 12, 33, 44, 2, 5, 6]`.
3. Write a Python program to count the odd and even numbers in the list `[10, 23, 11, 12, 33, 44, 2, 5, 6]`.
4. Write a Python program to interchange the first and last elements in the list `[10, 23, 11, 12, 33, 44, 2, 5, 6]`.
5. Write a Python program to duplicate all the items in the list `li = [1, 2, 3]`, such that the result is:
`output = [1, 2, 3, 1, 2, 3, 1, 2, 3]`.
6. Find the smallest element in the list `[10, 23, 11, 12, 33, 44, 2, 5, 6]`.
7. Find the greatest element in the list `[89, 23, 24, 2, 55, 54, 64]`.
8. Find the repetitive elements in the list `[1, 2, 3, 4, 56, 1, 22, 23, 33, 23, 56]`.
9. Remove all the odd elements from the list `[10, 23, 11, 12, 33, 44, 2, 5, 6]`.
10. Find all non-repetitive elements in the list `[1, 2, 3, 4, 56, 1, 22, 23, 33, 23, 56]`.
11. Write a Python program to duplicate all items in the list `l = [1, 2, 3]` to produce the result:
`result = [1, 2, 3, 1, 2, 3, 1, 2, 3]`.
12. Find the second greatest element in the list `[89, 23, 24, 2, 55, 54, 64]`.

13. Reverse the list `[1, 2, 3, 4, 56, 1, 22, 23, 33, 23, 56]`.
14. Arrange the list `[1, 2, 3, 4, 56, 1, 22, 23, 33, 23, 56]` in ascending order.
15. Arrange the list `[1, 2, 3, 4, 56, 1, 22, 23, 33, 23, 56]` in descending order.
16. Write a Python program to print all the vowels present in the given list of strings `["Dreamer", "infotech"]`.
17. Write a Python program to take input from the user to create a list of elements. The user should input each element of the list one by one. Create a list with these elements (maximum of 5 elements).
18. Write a Python program to generate a list of numbers in reverse order from 10 to 1 using list comprehension.
19. Create a list of square numbers from 1 to 10 using list comprehension.
20. Create a list of even numbers from 1 to 10 using list comprehension.
21. Filter strings from the list `language = ['python', 'php', 'java', 'c++', 'javascript', 'ruby']` that contain a specific letter provided by the user, using list comprehension.
22. Flatten a nested list like `[[1,2], [3,4], [5,6]]` into a single list.
23. Find the frequency of each element in a list.

List:

`[1, 2, 2, 3, 4, 1, 5, 2]`

Output:

`1 → 2 times`

`2 → 3 times`

`3 → 1 time`

4 → 1 time

5 → 1 time

24. Check whether a given list is a palindrome.

List:

[1, 2, 3, 2, 1]

Output:

Palindrome

25. Find the longest string in a list of strings.

List: ["cat", "elephant", "dog", "hippopotamus"]

Output: "hippopotamus"

26. Count how many strings in a list start with a vowel.

List:

["apple", "banana", "orange", "umbrella", "grape", "ice"]

Output:

4 strings start with a vowel

(apple, orange, umbrella, ice)

27. Replace all negative numbers in a list with zero

List:

[5, -3, 7, -1, 0, -8, 4]

Output:

[5, 0, 7, 0, 0, 0, 4]

28. Create a new list containing only prime numbers from a given list.

29. Convert a list of integers into a single integer number (e.g., [1, 2, 3, 4] → 1234).

30. Group elements of a list based on even and odd indices.

31. Find the largest even number in a list.

32. Find the majority element in a list (an element that appears more than $n/2$ times).

List:[2, 2, 1, 2, 3, 2, 2]

Output:2

33. Find the difference between the maximum and minimum values in a list.

List:[10, 4, 7, 2, 15, 6]

Output:13

34. Remove every third element from a list.

List:[1, 2, 3, 4, 5, 6, 7, 8, 9]

Output:[1, 2, 4, 5, 7, 8]

35. Insert an element at every even index of a list.

Original List:[1, 2, 3]

Element to Insert: 0

Output:[0, 1, 0, 2, 0, 3]

36. Rearrange a list so that positive and negative numbers alternate.

List: [1, 2, -3, -4, 5, -6]

Output:[1, -3, 2, -4, 5, -6]

37. Find **all pairs of elements** in a list whose sum equals a given target number.

List = [2, 4, 3, 5, 7, 8, 9]

Target = 7

Valid pairs:(2, 5), (4, 3)

38. Find the Missing Number from a Sequence.

List:[1, 2, 4, 5, 6]

Expected Range: 1 to 6

Output:3

39. Split a List into Chunks of Size 3.

List:[1, 2, 3, 4, 5, 6, 7, 8, 9]

Output:[[1, 2, 3], [4, 5, 6], [7, 8, 9]]

40. Reverse Each String in a List.

List:["python", "java", "script"]

Output: ["nohtyp", "avaj", "tpircs"]

41. Extract All Digits from a List of Strings

List:["abc123", "45def", "gh6"]

Output:["123", "45", "6"]

42. Find All Anagram Groups in a List of Words.

List:["eat", "tea", "tan", "ate", "nat", "bat"]

Output:[["eat", "tea", "ate"], ["tan", "nat"], ["bat"]]